

Curriculum Vitae

Personal Data

Name	Tamás Horváth
Date / Place of birth	05 November 1985, Budapest
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Familienstand	single



Professional Experience

10.2024 – 09.2025

Werkstudent, BMW AG, Munich

Responsibilities:

- Developed and optimized cell-contacting concepts and parametric CAD models in CATIA V5 under tight packaging, sheet-metal manufacturing and system constraints.
- Created CAD models to simplify FEM boundary-condition definition and improved the CAD-to-simulation interface, saving approximately 2–3 hours per development iteration.
- Developed Python-based automated evaluation tools for cell-contacting design concepts, including a computer-vision approach that fully automated a previously manual specification check.
- Supported validation-scope definition, independently solved CAD design tasks and proposed technical optimizations for future cell-contacting systems.

06.2021 – 05.2023

Entwicklungsingenieur, Hilite Germany GmbH, Nürtingen
Responsibilities:

- Designed and verified safety-critical bolted joints for hydraulic engine systems according to VDI 2230, considering customer requirements, thermo-mechanical loading, material behavior and manufacturing constraints.
- Established a cross-functional bolted-joint development process from concept to series production, including design reviews, simulation and test alignment, calculation tools, technical training and feedback into subsequent product development.

12.2018 – 10.2020

Konstrukteur, Schotstek GmbH Contracted at Hilite Germany GmbH, Nürtingen
Responsibilities:

- Developed and verified bolted-joint concepts according to VDI 2230, considering functional, material, manufacturing, economic and environmental requirements.
- Advanced numerical simulation models and test methods for bolted joints and supported drawing reviews, validation and series-assembly readiness.

05.2018 – 11.2018

Konstrukteur, Schotstek GmbH Contracted at Valeo GmbH, Bietigheim-Bissingen
Responsibilities:

- Developed CATIA V5 concepts for charge-air routing systems with integrated charge-air coolers and prepared drawings and tolerance calculations.
- Supported customer presentations and collaborated with prototype build, simulation and production planning teams on project documentation and design coordination.

07.2016 – 04.2018

Konstrukteur, Schotstek GmbH Contracted at MAHLE Filtration GmbH, Stuttgart

Responsibilities:

- Developed and evaluated mechanical concepts for air filters and high- and low-pressure air systems in CATIA V5, considering functional, material, manufacturing and economic requirements.
- Created 3D designs, drawings and tolerance analyses, coordinated solutions with adjacent engineering functions and automated recurring engineering workflows using VBScript and CATScript.

07.2014 – 06.2016

Konstrukteur, Hays Temp GmbH Contracted at Semcon Bad Friedrichshall GmbH, Bad Friedrichshall

Responsibilities:

- Developed series-production intake and charge-air-cooler systems using skeleton-based parametric CAD architectures in Siemens NX and CATIA, integrating packaging, manufacturing, assembly and service requirements.
- Designed injection-molded housings and internal flow geometry by incorporating flow, thermal and FEM results, including flow-guiding features that resolved conflicts between airflow targets, throttle-body orientation and available package space.
- Structured a common flow-space CAD model so geometry changes propagated consistently into upper and lower shells, supporting parallel development and efficient design iterations through series maturity.
- Performed package studies, kinematic simulations and FEM assessments and coordinated design solutions with simulation, customers, suppliers and adjacent engineering disciplines.

Education

10.2020 – 09.2025	M.Sc. Mechanical Engineering, Technische Universität München, Munich ■ Course of study: Engineering, Production, and Management in Mechanical Engineering (Engineering) ■ Main subject: Structural Mechanics & FEM, Engineering Design & Manufacturing, System Modeling & Simulation ■ Thesis: Feature Extraction Methods for Improving Machine Learning Detection of Knee Osteoarthritis Using Vibroarthrography Signals ■ Term paper: Data Preprocessing Techniques for Machine Learning Detection of Knee Osteoarthritis Using Vibroarthrography Signals ■ Research Practice: LSTM model predicting positional failures in a 6-DoF industrial robot using open-source ML tools. ■ Research Practice: Designed energy-efficient trajectory planning algorithms for UAVs using C++ and ROS.
09.2005 – 02.2012	B.Sc. Mechanical Engineering, Budapesti Műszaki és Gazdaságtudományi Egyetem, Budapest ■ Course of study: Mechanical Engineering (R&D Engineering) ■ Main subject: Mechanical Design & Materials, CAD/FEM & Computational Methods, Manufacturing & Production Engineering ■ Thesis: Designing a mechanism for kinematics take-up of human and robotic fingers with rehabilitation purposes ■ Research Practice: Designed a Human-Machine Interface (HMI) 6-Degree-of-Freedom Robotic System for Physiotherapy Purposes
09.2000 – 07.2004	Érettségi (Hungarian University Entrance Qualification), Petőfi Sándor Gimnázium, Budapest ■ Main subject: Mathematics, History

publications

02.2026 – present	Matrix Profile-Based Multiclass Cartilage Condition Assessment from Vibroarthrography Signals in preparation
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extra-curricular

11.2025 – present	Volunteer, Ludwig und Kunst, Munich Assisting with event organization, logistics, and operational management.
07.2023 – 08.2023	Intern, Musik Sommer, Munich Responsible for event coordination, on-site management, and technical assistance.

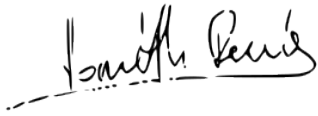
interests

Programming	Ongoing development of programming skills through personal projects using REST APIs, SQL, and AI-assisted development (Copilot, Claude CLI, Ollama for local LLMs).
Cultural Engagement	Actively engaged in music and art events; organizational support, community building, and networking.
Sport	Enjoy outdoor sports such as running, skiing, and bicycling.

Skills

CAD & CAE Tools	CATIA V5	Expert
	Siemens NX	Expert
	CREO Parametric	Basic
	ANSYS	Intermediate
PLM & Version Control	PRISMA	Expert
	Windchill	Advanced
	SAP PLM	Intermediate
	Git	Intermediate
Programming & Automation	AutomaPython	Advanced
	MATLAB/Simulink	Advanced
	VBA	Intermediate
	C/C++	Advanced

Munich, 09 August 2026

A handwritten signature in black ink, appearing to read "Sarah Rein". The signature is written in a cursive style with a horizontal line underneath the name.